computational game theory

com810s

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games of incomplete information

intro

- price competition games
  - players (firms) know their cost, but not their opponents' cost
- negotiating parties may have different beliefs about the context
- auctions
  - bidders don't know competition's valuation

- players do not have **common knowledge** of the game
- knowledge
  - the players
  - payoffs
  - possible strategies
  - what other players know

- consider the game with two players, P_1 and P_2
- P_2 has two options: option 1 and option 2

option 1

$$M_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}; \quad N_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

option 2

$$M_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}; \quad N_2 = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

- P_1 is uncertain about which game they are playing
- P_1 believes there is a chance (2/3) that option 1 is playing

incomplete information concepts

types

- players can have a type
  - P_2 has two types in the previous example

T_ι : player P_ι type space

$t_{\iota 1} \in \mathsf{T}_\iota$ possible type for Player P_ι

payoffs

$$u_l : A \times T_l \to \mathbb{R}$$

$$u_l(a_1, \ldots, a_\ell; t_l)$$

- note
  - could generalise the payoff to more types
    - action depends on other player's types
  - can set the payoff to minus infinity to block the action
  - type spaces and all possible games are known, but not the current game to all players

Beliefs

- probability distribution over other players' types
  - a player can have belief over other players's type

$$\mathrm{p}(\mathrm{t}_j \mid \mathrm{t}_i) \in [0,1]$$

$$\sum_{\mathrm{t}_j \in \mathrm{T}_{-i}} p(\mathrm{t}_j \mid \mathrm{t}_i) = 1$$

- note
  - the joint probability distribution over types is known to all players
  - only player P_i knows about its type

$$p(\mathrm{t}_J \mid \mathrm{t}_l) = \frac{p(\mathrm{t}_J, \mathrm{t}_J)}{p(\mathrm{t}_l)}$$

Bayes rule

Bayesian games

$$\mathscr{G} = \langle \mathsf{P}, \mathsf{A}, \mathsf{T}, \mathsf{B}, \mathbf{u} \rangle$$

- players
- actions
- types
- beliefs
- payoffs

$$\mathscr{G} = \langle \mathsf{P}, \mathsf{A}, \mathsf{T}, \mathsf{B}, \mathbf{u} \rangle$$

- example on slide 6
  - two players (P_1 and P_2)
  - two actions per player
  - P_2 has tow types (t_{21} and t_{22})
  - P_1's beliefs
  - payoffs as indicated in the matrices

$$p_1(\mathrm{t}_{21} \,|\, \mathrm{t}_1) = \frac{2}{3}; \quad p_1(\mathrm{t}_{22} \,|\, \mathrm{t}_1) = \frac{1}{3}$$

$$\mathsf{M}_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}; \quad \mathsf{N}_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\mathsf{M}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}; \quad \mathsf{N}_2 = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

Bayesian Nash equilibrium

- strategy
  - different types of players may choose different actions
- payoff stages
  - ex ante: no knowledge about types
  - interim: a player knows its type, but no information about other players
  - ex post: a player knows all players' types
- incomplete information is interim

- expected payoff

$$\mathrm{E}_{\iota}(\mathrm{S}_{\iota}) = p(\mathrm{t}_{J} \mid \mathrm{t}_{\iota}) \times \mathrm{u}_{\iota}(\mathrm{S}_{\iota}, \mathrm{S}_{-\iota}; \mathrm{t}_{\iota}, \mathrm{t}_{-\iota})$$

- Bayesian Nash equilibrium
  - strategy profile (s^*_1, ...s^*_n) is a pure strategy Bayesian NE if

$$S_i^*(t_i) \in argmax \sum_{t_i} p(t_j \mid t_i) \times u_i(S_i, S_{-i}; t_i, t_{-i})$$

$$\mathscr{G} = \langle \mathsf{P}, \mathsf{A}, \mathsf{T}, \mathsf{B}, \mathbf{u} \rangle$$

$$\mathsf{A}_1 = \{a_1, a_2\}; \quad \mathsf{A}_2 = \{b_1, b_2\}$$

$$p_1(t_{21} \mid t_1) = q; \quad p_1(t_{22} \mid t_1) = 1 - q$$

$$\mathsf{M}_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}; \quad \mathsf{N}_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\mathsf{M}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}; \quad \mathsf{N}_2 = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

- example
  - two players (P_1 and P_2)
  - two actions per player
  - P_2 has tow types (t_{21} and t_{22})
  - P_1's beliefs
  - payoffs as indicated in the matrices

- P_2 knows which game is being played currently
- P_1 needs to form expectations about the game outcome

- steps
  - define all strategies for P_2
  - compute the expected payoffs for P_1
  - find the best responses for both players and determine the equilibrium
  - see illustration

$$\mathscr{G} = \langle \mathsf{P}, \mathsf{A}, \mathsf{T}, \mathsf{B}, \mathbf{u} \rangle$$

$$\mathsf{A}_1 = \{a_1, a_2\}; \quad \mathsf{A}_2 = \{b_1, b_2\}$$

- example
  - two players (P_1 and P_2)
  - two actions per player
  - P_2 has tow types (t_{21} and t_{22})
  - P_1's beliefs
  - payoffs as indicated in the matrices

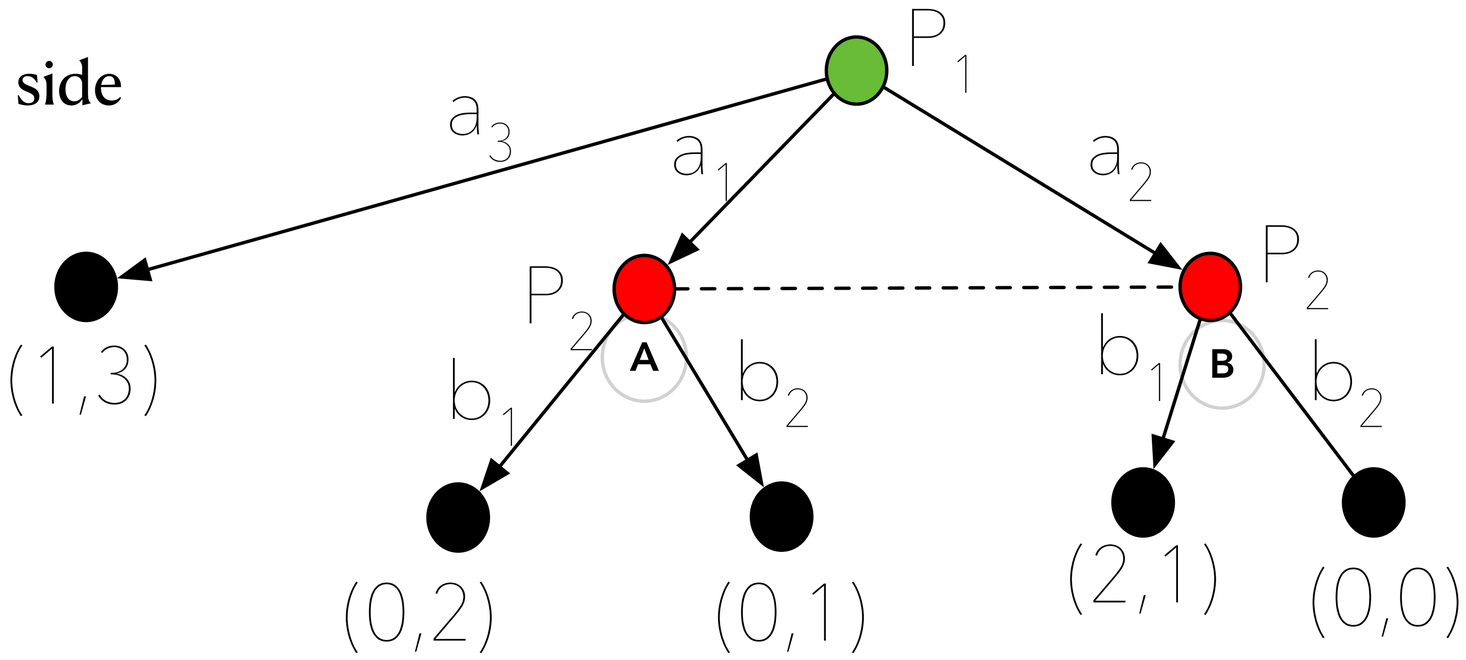
$$p_1(t_{21} \mid t_1) = \frac{1}{2}; \quad p_1(t_{22} \mid t_1) = \frac{1}{2}$$

$$\mathsf{M}_1 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}; \quad \mathsf{N}_1 = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\mathsf{M}_2 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}; \quad \mathsf{N}_2 = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix}$$

dynamic games

- consider the game on right-hand side
  - are there any NE?
  - if yes, are they rational?



- previous game has two PSNE: (a_3, b_2) and (a_2, b_1)
- however, the first one is not rational
  - b_2 is not a credible threat
- refine the NE with SPNE

- back to belief system

Let μ represent the belief system at node A

$$\mathsf{E}_2(\mathrm{b}_1) = 2\mu + (1 - \mu) = \mu + 1$$

$$\mathsf{E}_2(\mathrm{b}_2) = \mu$$

- b_1 is preferable to b_2

- belief requirements
  - well-defined beliefs at information set
  - beliefs can depend on chance (nature) player or strategies
  - players strategies must be sequentially rational
  - when an information set is reached with strictly positive belief, it is on the equilibrium path

- Perfect Bayesian NE
  - BNE with a consistent belief system

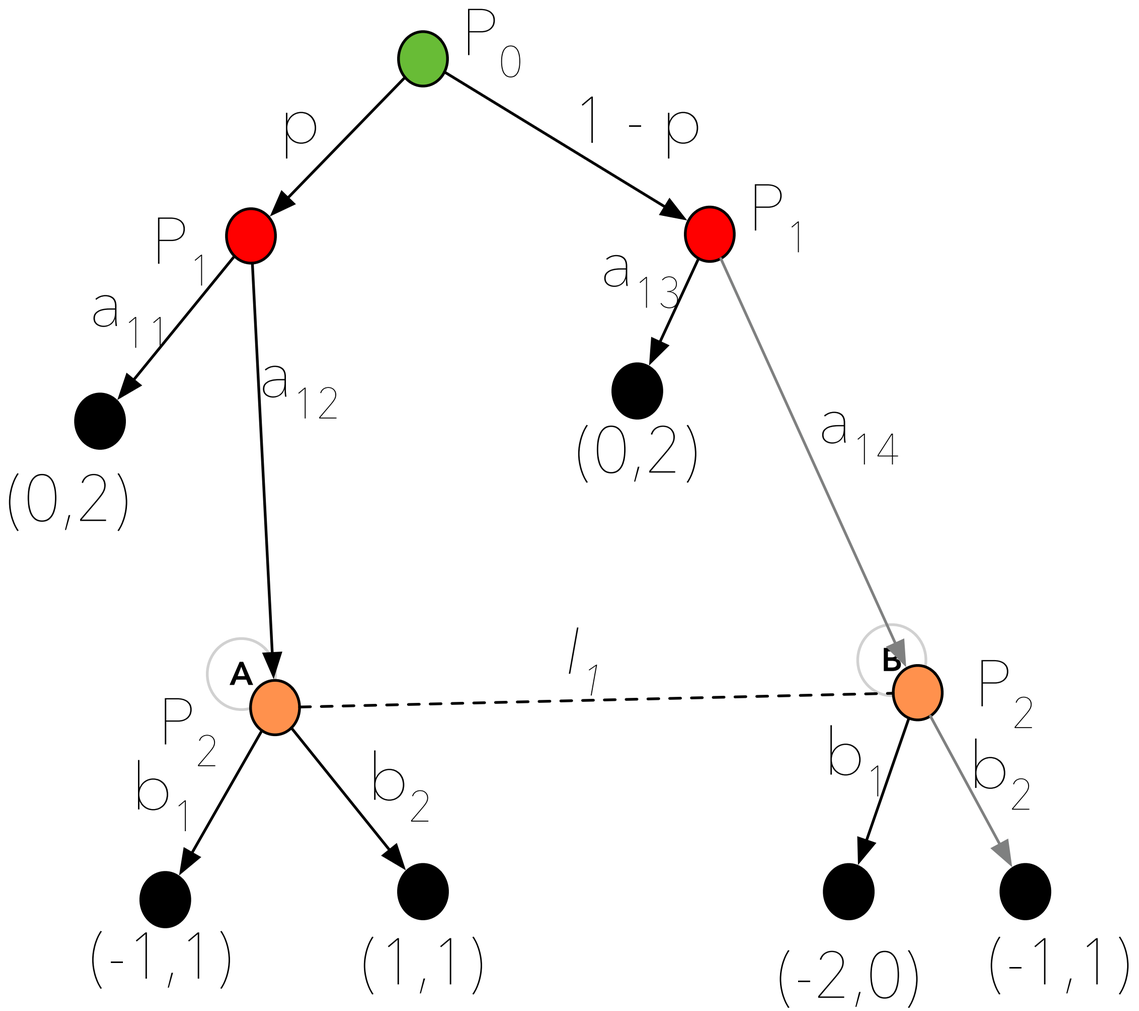
- back to the example
  - (a_2, b_1)
    - information set is on the equilibrium path
    - b_1 dominates b_2
    - P_1 prefers a_2 to a_3
    - belief probability at node A must be 0

- back to the example
  - previous game has two PSNE: (a_3, b_2) and
  - (a_3, b_2)
    - information set is off the equilibrium path
    - we can assign any belief probability
    - Slide 35 shows that b_1 dominates b_2
    - (a_3, b_2) is not sequenally rational

$((a_2, b_1); \mu = 0)$ is the PBNE

- note the similarity with sequential equilibrium

- consider the game on right-hand side
  - find the PBNE



summary

- static games of imperfect information
  - BNE
- dynamic games
  - PBNE